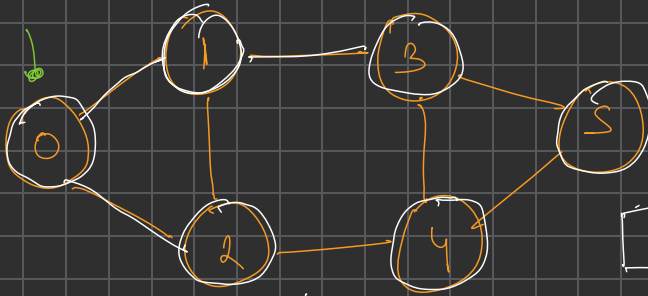




$$\begin{array}{r}
 0 - 4 = 2 \\
 4 - 5 = 1 \\
 \hline
 0 - 5 = 3
 \end{array}$$

Undirected, weighted graph

$S \rightarrow 4$



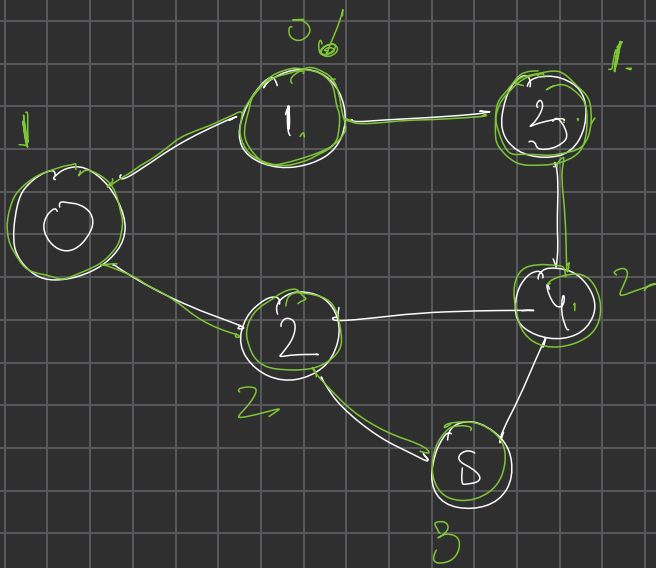
queue

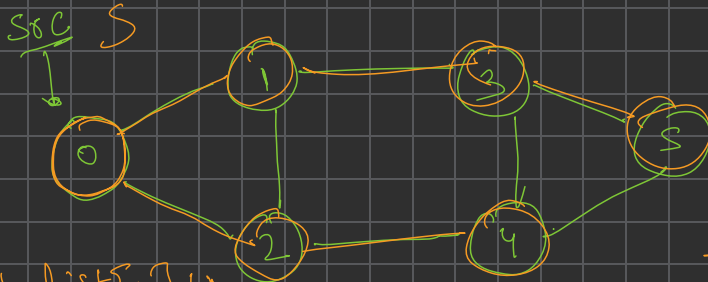
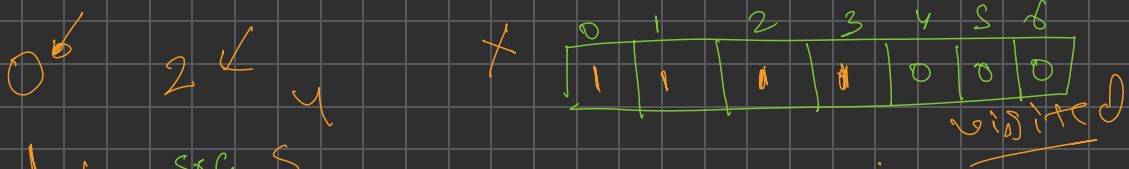


dist

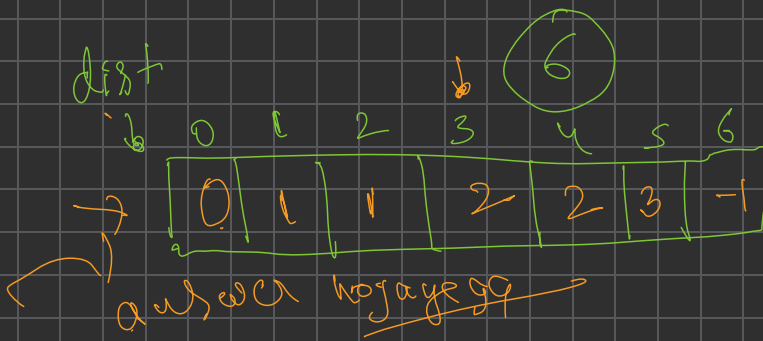
0	1	2	3	4	5
0	1	1	2	2	3

$\uparrow$
 add $\begin{array}{l} 0 - 3 \Rightarrow 2 \\ 3 - 5 = 1 \\ 0 - 5 \Rightarrow 3 \end{array}$





$$\text{dist}[1] = \text{dist}[0] + 1$$

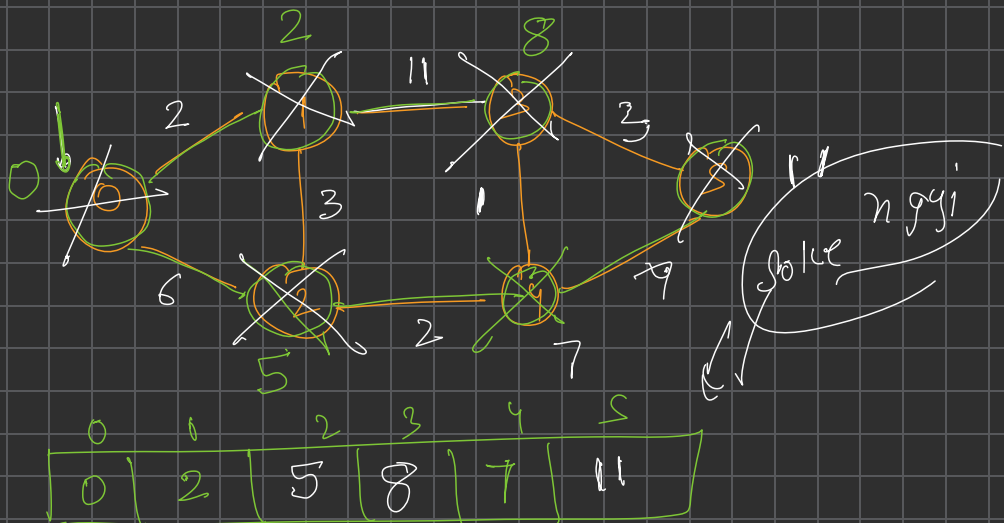


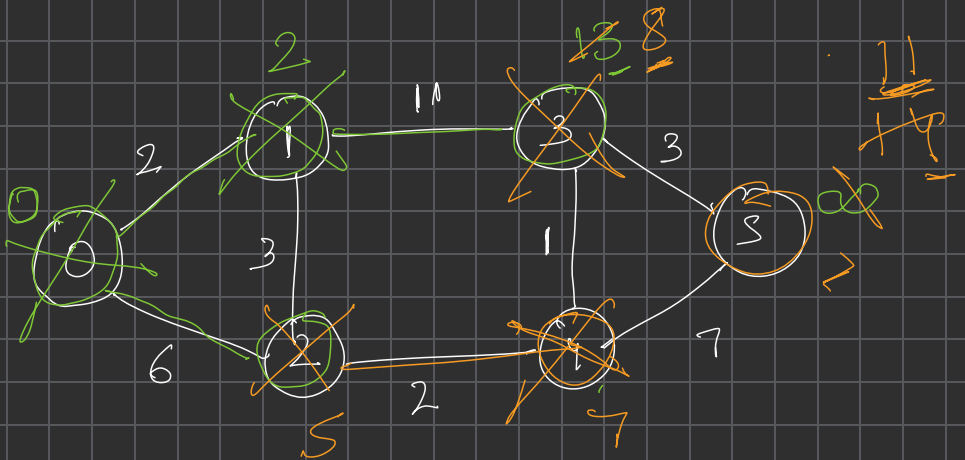
- ① dist array create karo of size V & initialize -1,
- ② src node ko queue me push karo,
 $\text{dist}[\text{src}] = 0$;

(3)

while (!q.empty())
{
 (1) front node',
 (2) Look at all the neighbours
 of that node
 ↳ if neib dist = -1 (unvisited)
 dist[neib] = dist[node] + 1;
 push neib into queue;
}

$$\text{dist}[s] = 4(3)$$



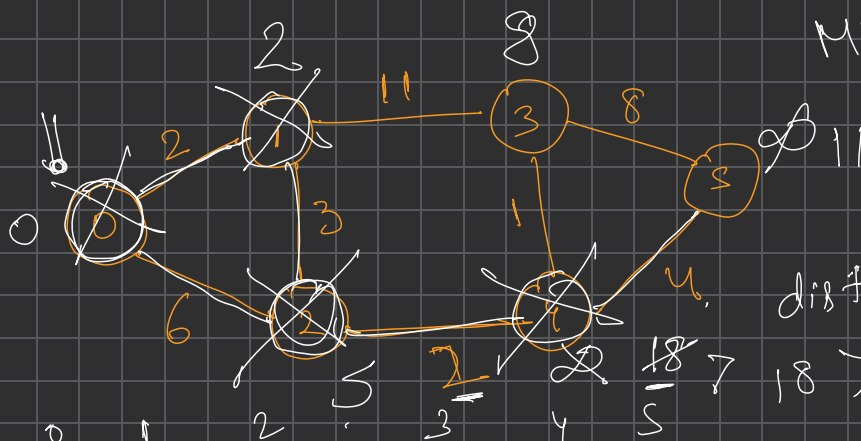


0	1	2	3	4	5
0	2	5	8	7	11

213, 32

problem
solve
hgyi

pg. 1
Min heap



dist[u]
18 > 5+2

0	1	2	3	4	5
0	2	5	8	7	11

① dist array v , initlize INT_Max ;

(2) $\text{dist}[\text{src}] = 0$

(2) dist $\in \mathbb{R}$

(3) min-heap $\Rightarrow$ pair $\langle \text{int}, \text{int} \rangle$

dist $\uparrow$ node $\rightarrow$ h

(2) $pa \leftarrow pa \cdot \text{push} \{ 0, \text{src} \}$

```
while (!pq.empty())
```

{ node = element
}

```

node = prev;
if (dist[node] < pq.dist)
    continue; (skip);

```

Look at all the neighbors of the node.

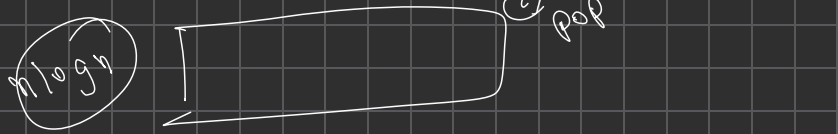
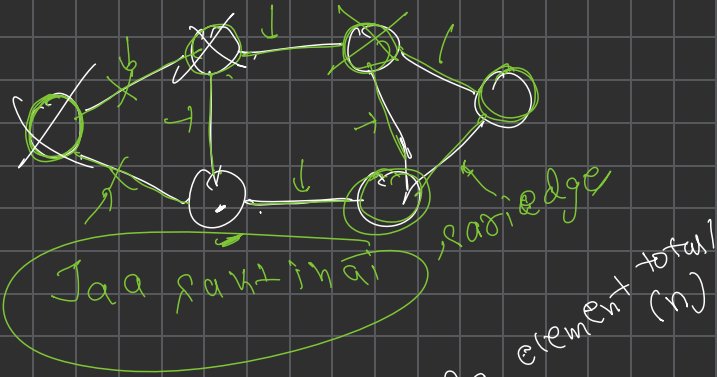
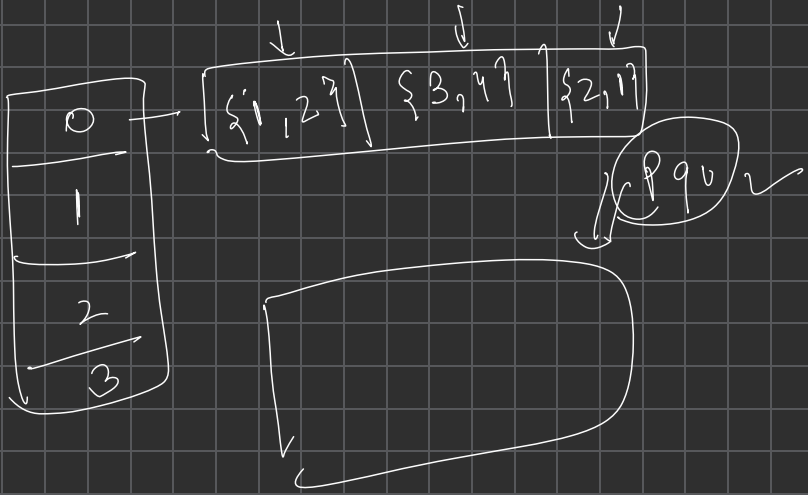
node:

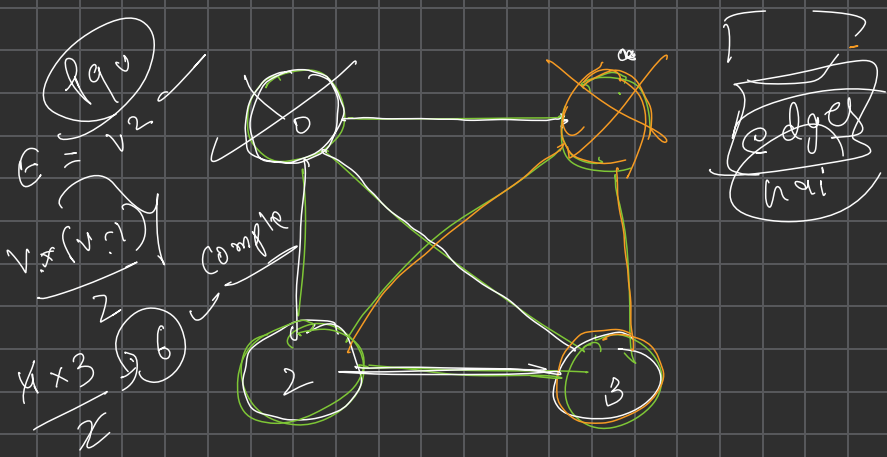
if ($\text{dist}[\text{neib}] > \text{dist}[\text{node}] + \text{weight}$)

$\text{dist}[\text{neib}] = \text{dist}[\text{node}] + \text{weight}$

$\{$
 $\text{dist}[\text{neib}] = \text{dist}[\text{node}] + \text{weight}$
 $\text{pq.push}(\text{dist}[\text{neib}], \text{neib})$

↙ ↘





$n \log n$
 $E \log E$
 $E \Rightarrow v^2$

$\Rightarrow E \log v^2$
 $\Rightarrow 2E \log v$
 $\Rightarrow \sqrt{E \log v}$